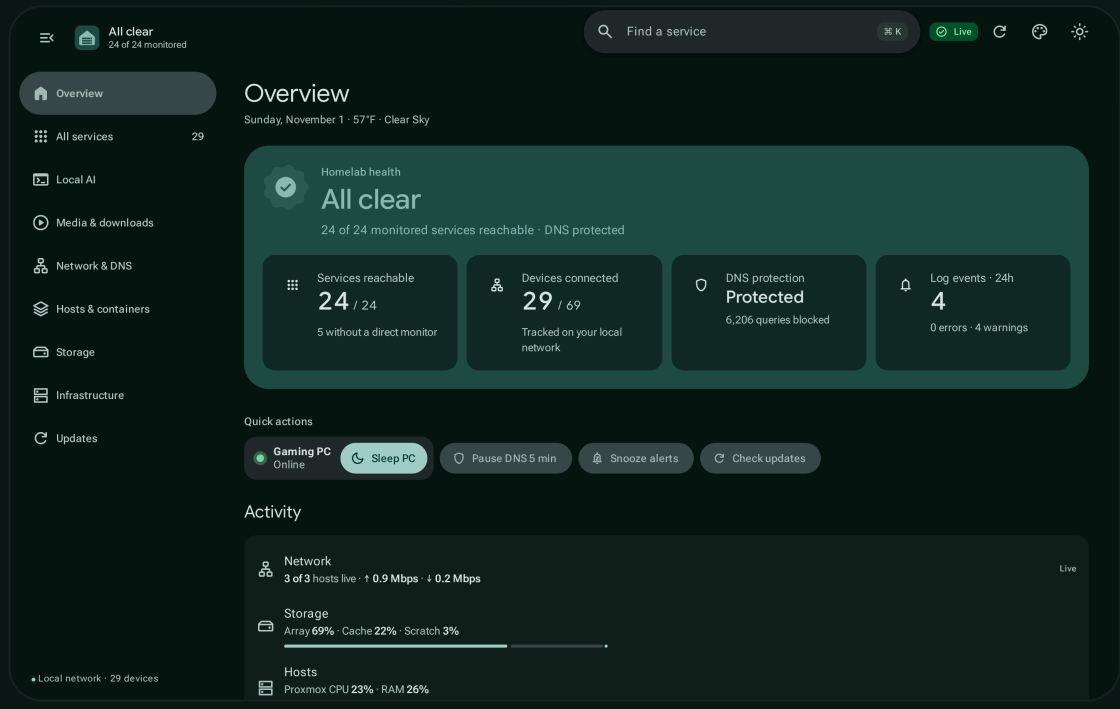
HOMELAB · NOVEMBER 2026

Homelab Command Center

A dashboard for my Proxmox and Unraid homelab. The first screen says whether anything needs attention. The other pages cover services, disks, network, backups, energy, updates and the local AI models.

React 18 · Python backend · Material 3 Expressive · Proxmox VE · Unraid · Intel Arc A310 · llama.cpp · Strata



29
services in the catalog

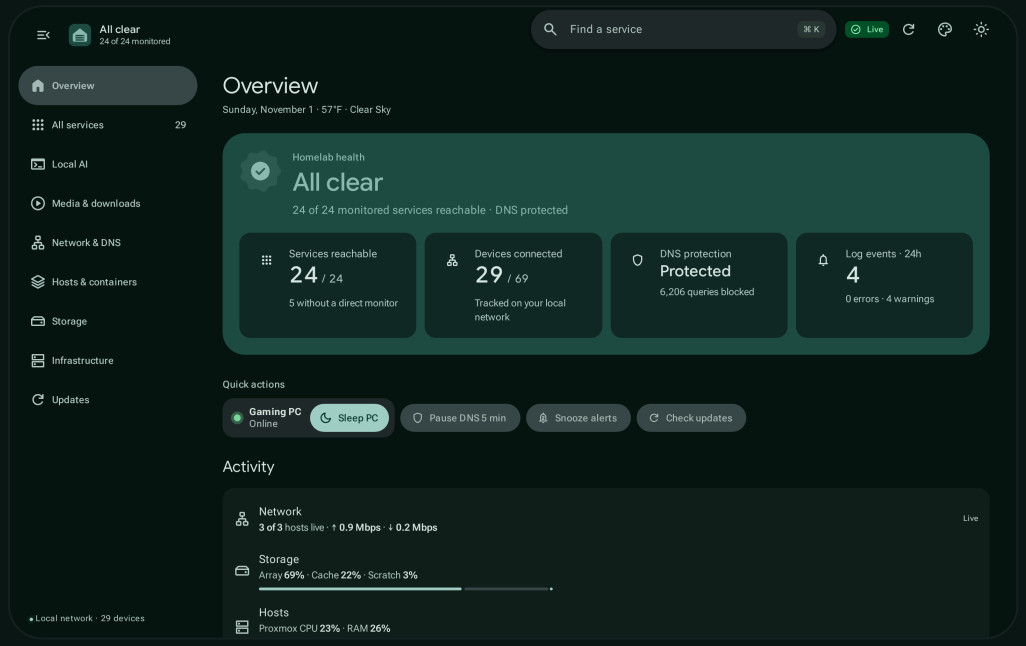
23
VMs, containers & apps

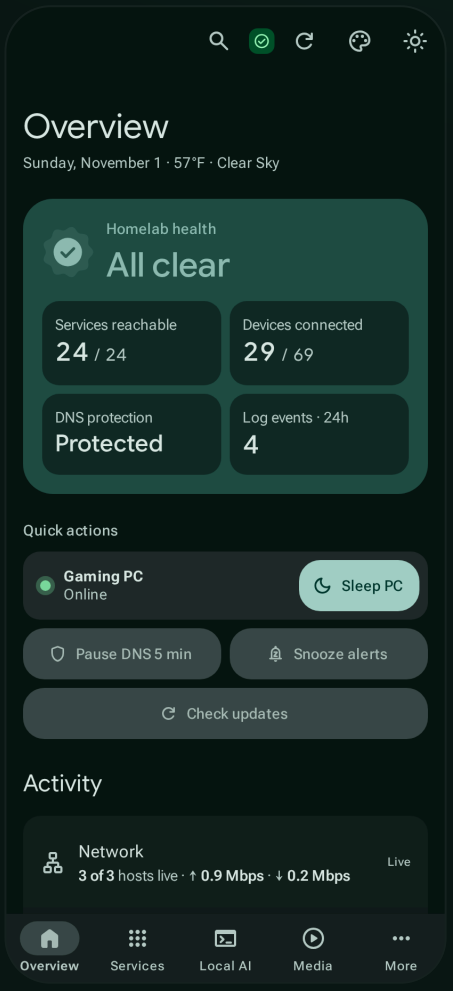
15
automated UI check suites

01 · OVERVIEW

Overview

The top card says "All clear" or lists what needs attention, and each item links to its page. Quick actions come next (wake or sleep the gaming PC, pause DNS blocking for five minutes, snooze alerts, check for updates), then live network, storage and host activity, the pinned services, and an Ask box at the bottom.





On a phone

The health card, quick actions and the start of the activity card fit on the first screen.

Confirmations

Actions that change something ask first, and the server only accepts them from the dashboard.

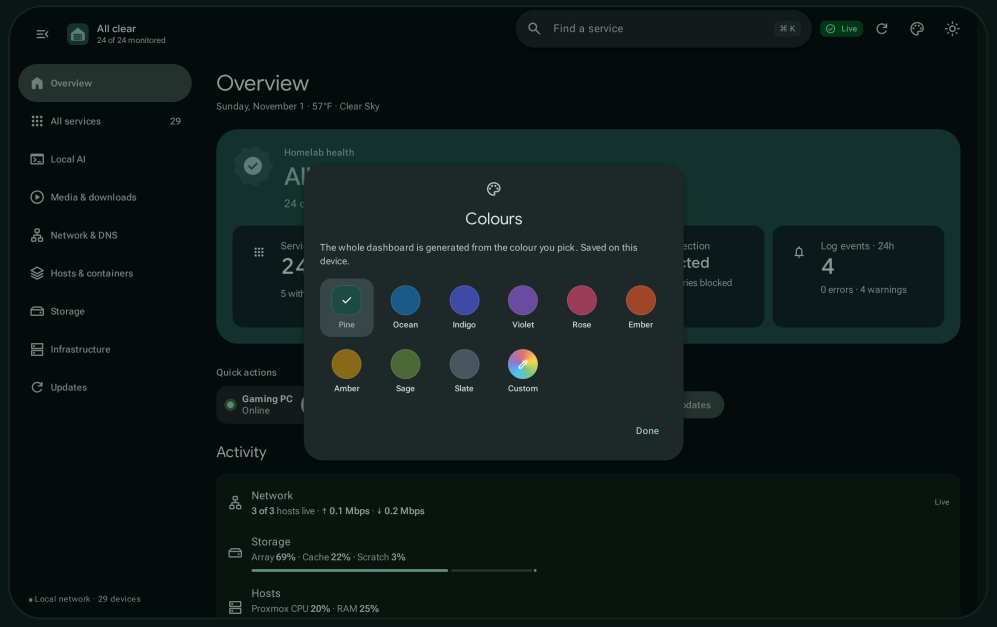
Missing readings

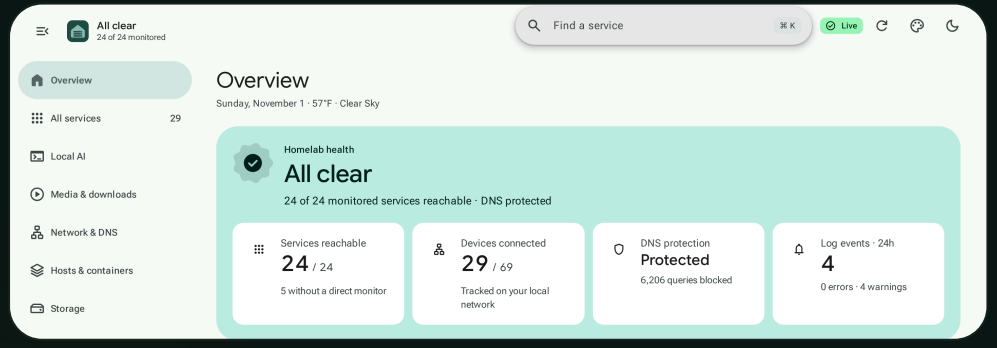
Each reading has a timestamp. If a source stops answering, its tile shows a dash instead of zero.

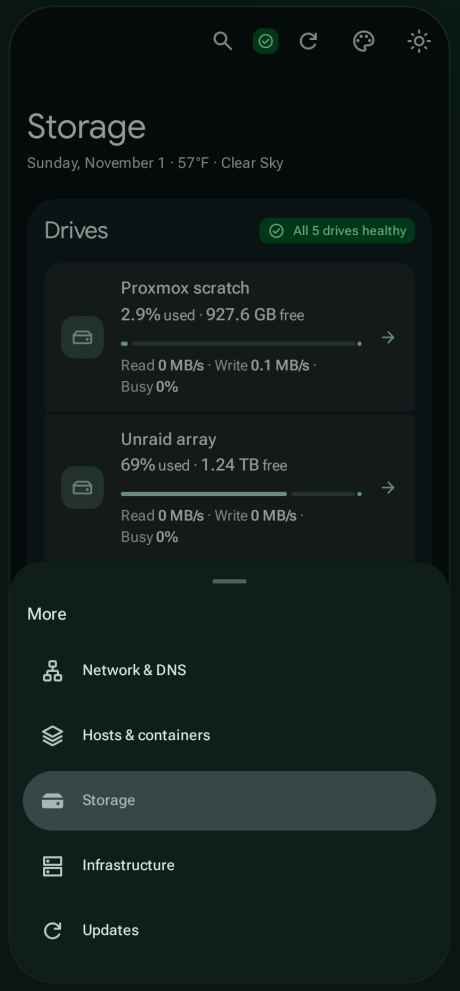
02 · DESIGN

Material 3 Expressive design

The colours come from Google's Material colour library, generated from one seed colour for both the light and dark themes. The seed is pine green by default; the Colours picker offers presets or any colour, and in the Android app it follows the phone's wallpaper like other Material You apps. Layout, shapes and motion follow Material 3 Expressive, Google's 2025 update to Material Design.







Shapes

The health icon sits in a scalloped shape that changes when something is wrong. Each service category has its own icon shape, and the loading indicator morphs between shapes.

Motion

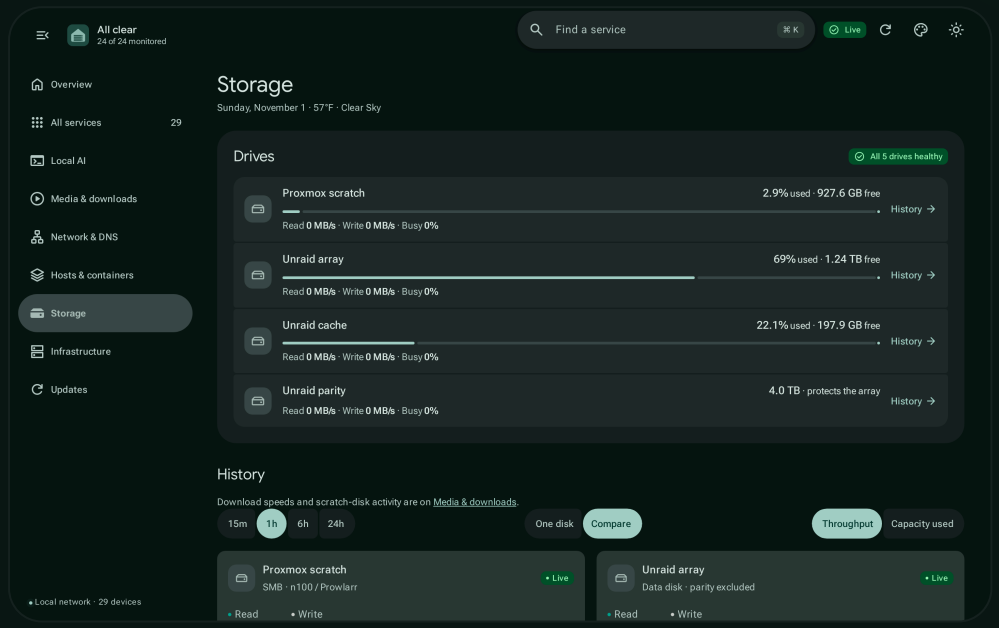
Pages slide into place, a tapped row turns into the next page's title, and buttons change shape when pressed. The system's Reduce motion setting turns all of it off.

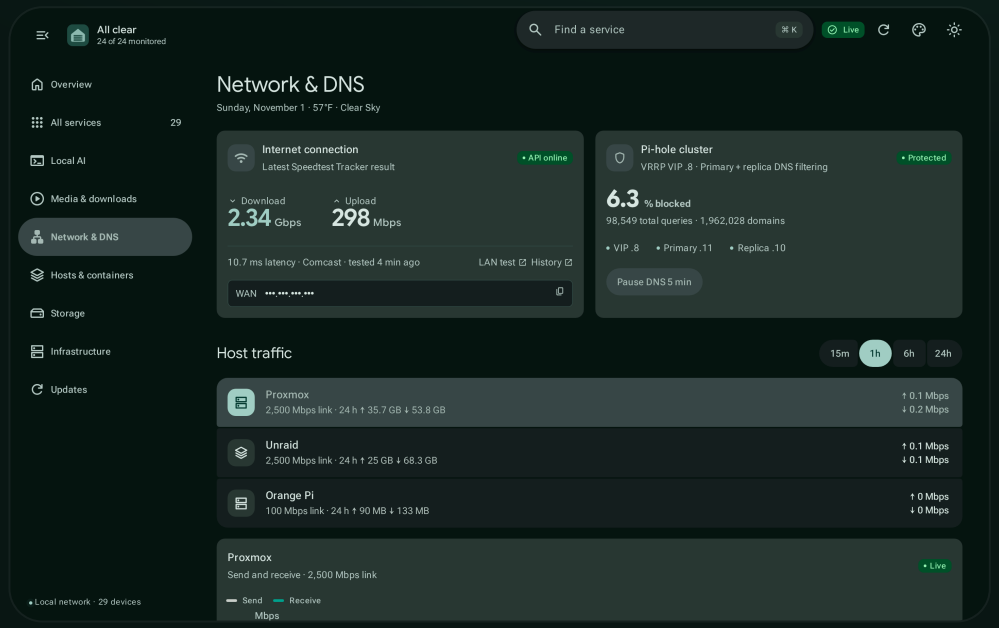
Accessibility

Text contrast is at least 4.5:1 in both themes. Automated tests cover keyboard focus, dialogs and navigation.

Storage and network

Storage opens with each drive's free space, live read and write rates and SMART health. Network opens with the latest speed test and the two Pi-hole servers. History charts come after that, one disk or host at a time on a phone or side by side on a wider screen.





Cache forecast

Estimates when the Unraid cache will fill and starts the mover before it does.

Chart colours

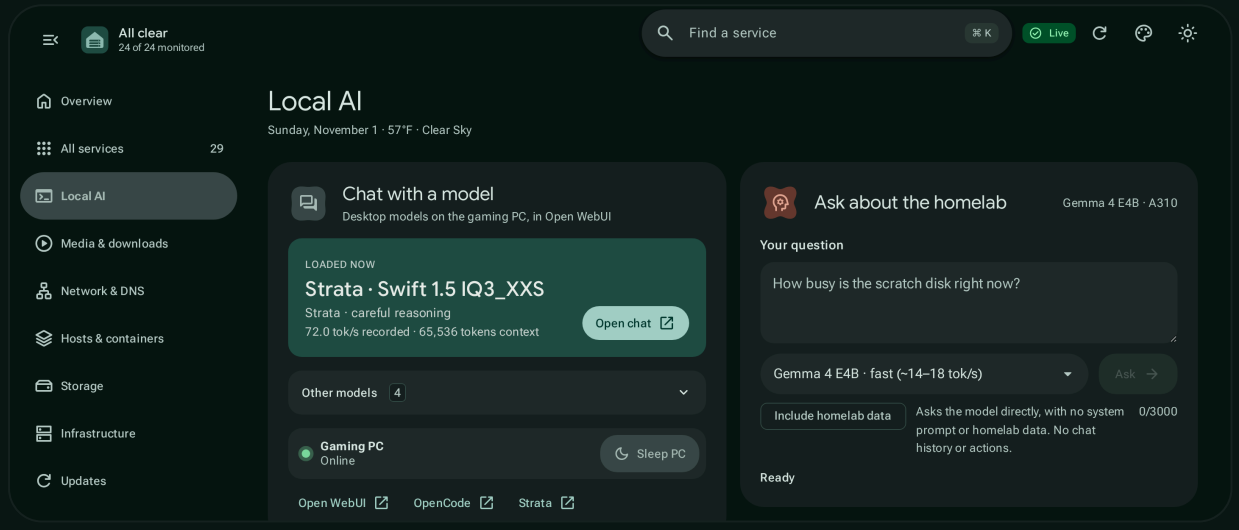
Each line colour was checked for colour-blind readers in both themes.

Gaps

If a host reboots, its chart shows a gap for the missing time.

Local AI

Four models run on the gaming PC's RTX 5070 Ti through llama.cpp, plus Strata, a 177B mixture-of-experts model streamed from the NVMe drive. Only one fits on the GPU at a time, so a small broker swaps them automatically when a request arrives. The page puts the loaded model first, and every request is logged with its swap time and speeds. A smaller Gemma model on the Arc A310 runs the nightly log review.





Recorded speeds

Gemma 4 26B · 168 tok/s
Strata · Swift 1.5 177B MoE · 72 tok/s
Qwen 3.6 35B MoE · 82 tok/s
Qwen 3.8 27B · 55 tok/s
Qwen 3.8 27B uncensored · 51 tok/s
Gemma 4 E4B on the A310 · 14 to 18 tok/s

GPU broker

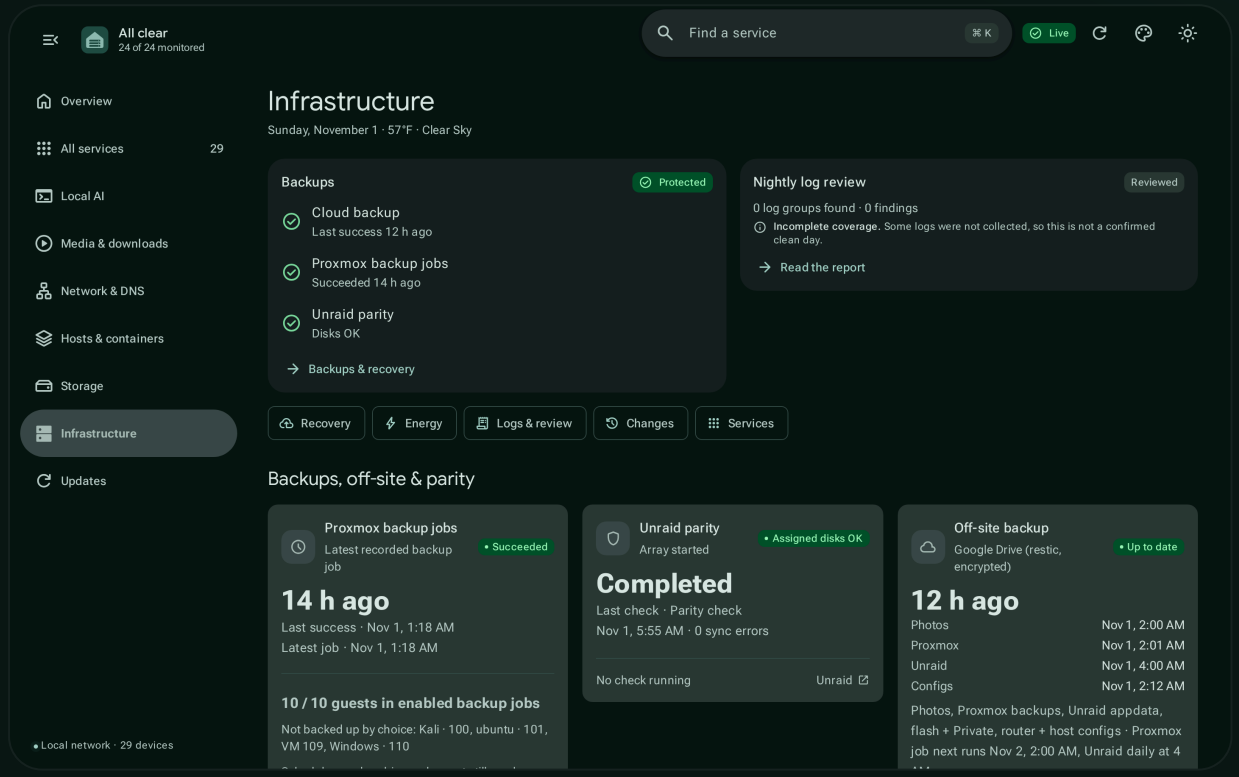
Every client talks to one address. A request for the other engine unloads the current one first, and nothing loads while a game is running.

Ask box

Sends just the question by default; a switch adds homelab readings. Strata answers can search the web.

Backups and recovery

Local Proxmox backups, the off-site copy in Google Drive and Unraid parity are shown together, with restore test results below them. Each night after the backups finish, the A310 model reviews the host warning logs.



Off-site copy

restic encrypts photos, Proxmox backups, Unraid app data and router and host configs before uploading them to Google Drive. It keeps 7 daily, 4 weekly and 12 monthly snapshots.

Restore tests

A scheduled test downloads sample files from Google Drive and compares them with the originals.

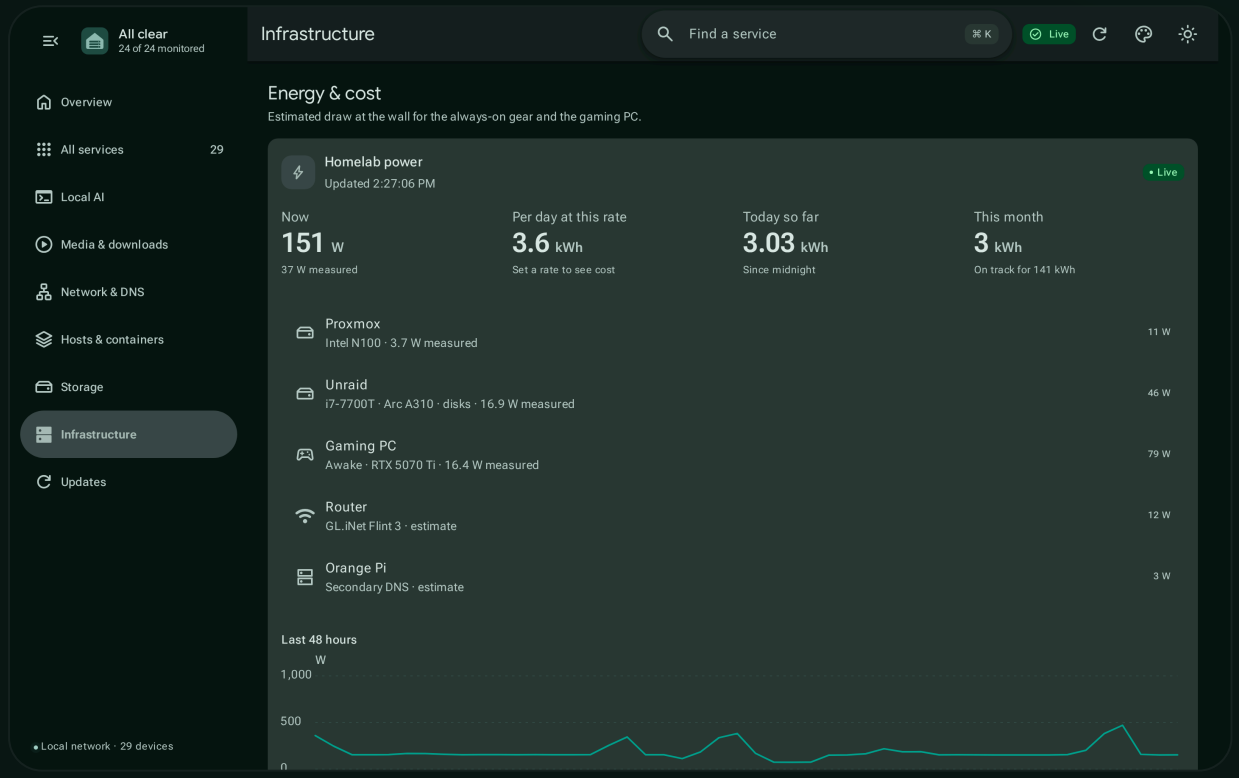
Section links

Buttons at the top of the page jump to recovery, logs, the change history or the service list.

06 · ENERGY

Energy

Power for the always-on gear and the gaming PC: CPU energy counters on Proxmox and Unraid, the Arc A310's own counter and the RTX 5070 Ti's board power are measured; motherboards, disks, the router and power-supply losses are labelled estimates. Hourly totals are kept for 400 days.



Measured or estimated

Every device says which it is, and the explainer lists each part's share. Totals are treated as ±20%.

Cost

With a private electricity rate set on the server, the tiles show cost per day, today and the month's projection. It is left out of these screenshots.

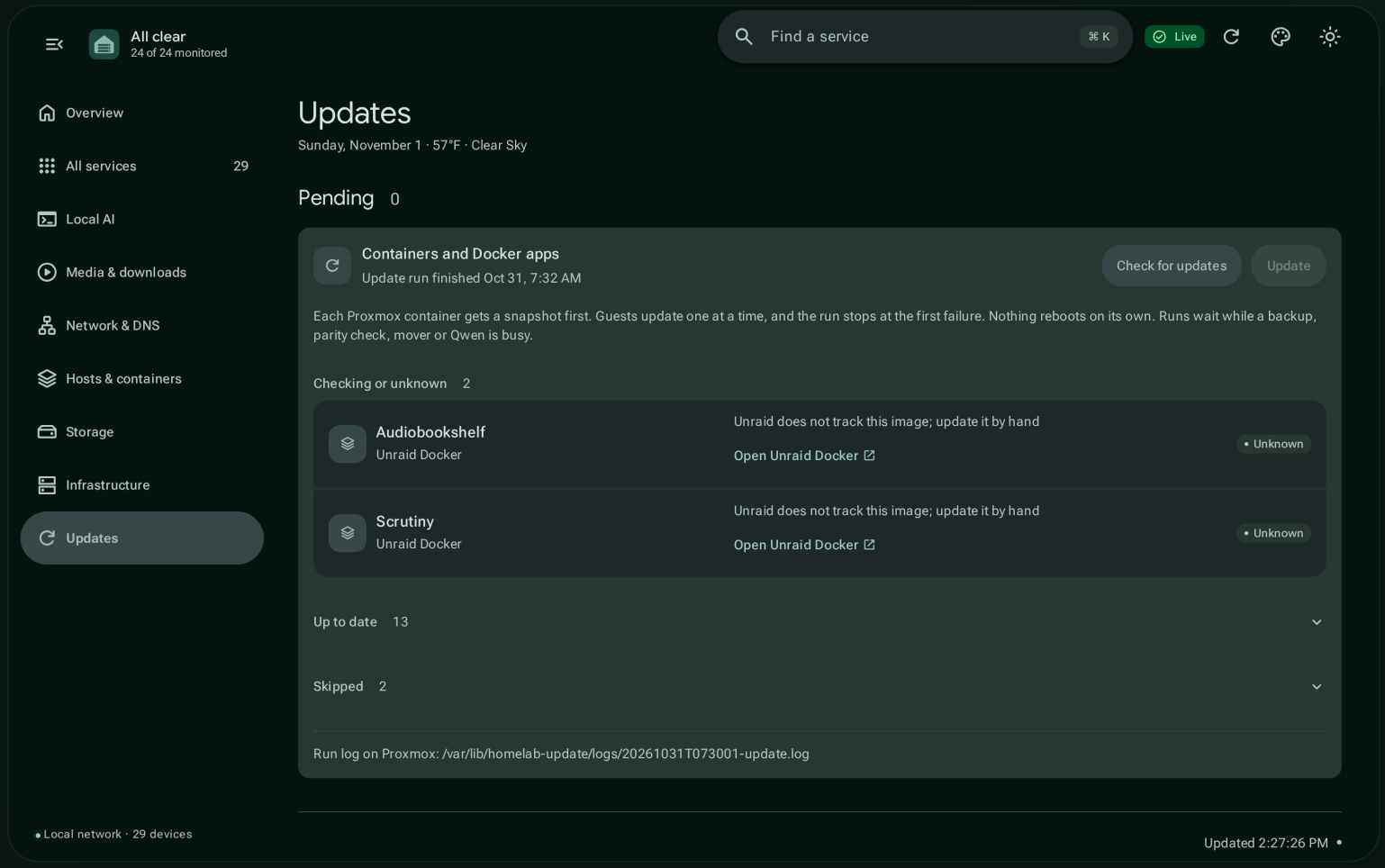
Morning digest

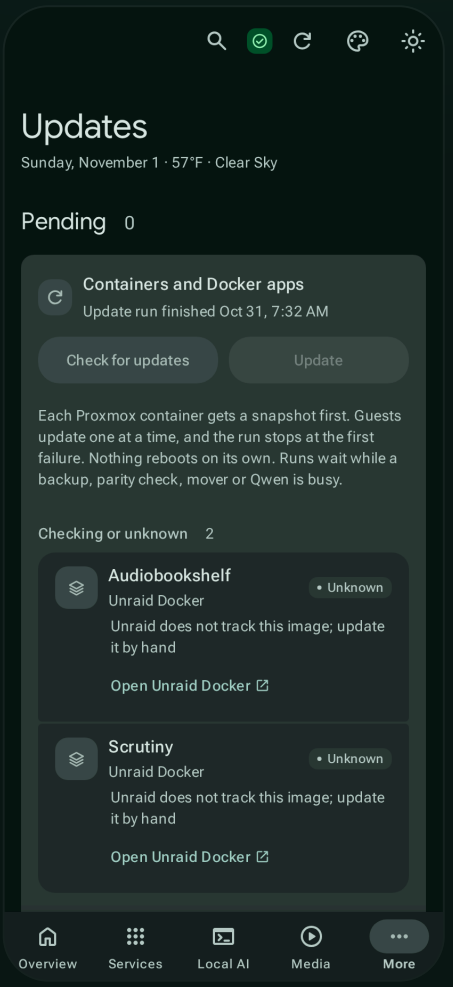
Yesterday's energy use is included in the 7 AM summary notification.

07 · UPDATES

Updates

Each Proxmox container is snapshotted before `apt upgrade` runs, and Docker apps update through Unraid's own updater. Guests update one at a time; a failure stops the run and shows the rollback command. Items with an update waiting are listed first, and current or skipped ones are folded away.





Hosts, containers and services

Proxmox, Unraid and the Orange Pi are at the top, followed by CPU, memory and state for every VM, container and Docker app. The services page groups everything by category, and unreachable, stale and unmonitored services are labelled in text as well as colour.

